

01. Introduction

Good nutrition is an important part of cancer treatment and recovery. A healthy diet helps the body fight infections, repair damaged tissues, maintain strength, and improve overall quality of life.

Cancer and its treatments may affect appetite, digestion, and the body's ability to absorb nutrients. Therefore, patients should follow a balanced and nutritious diet throughout treatment.

02. Importance of Nutrition During Cancer Treatment

Proper nutrition helps to:

- Maintain a healthy body weight
 - Increase energy levels
 - Improve immune function
 - Support tissue healing
 - Reduce treatment-related side effects
 - Improve tolerance to chemotherapy and radiation therapy
 - Enhance recovery after surgery
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03. Essential Nutrients

The body requires different nutrients for proper functioning.

Carbohydrates

Carbohydrates provide energy needed for daily activities and healing.

Healthy sources include:

- Brown rice
 - Whole wheat bread
 - Oats
 - Sweet potatoes
 - Whole grains
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Proteins

Protein is essential for repairing tissues and maintaining muscle mass.

Good protein sources include:

- Eggs

- Fish
- Chicken
- Lean meat
- Milk
- Yogurt
- Paneer
- Lentils
- Beans
- Soy products

Patients undergoing treatment generally require more protein than usual.

Healthy Fats

Healthy fats provide energy and support cell growth.

Healthy sources include:

- Nuts
- Seeds
- Olive oil
- Avocados
- Peanut butter

Limit foods containing trans fats and excessive saturated fats.

Vitamins and Minerals

Vitamins and minerals help support immunity and normal body functions.

Eat a variety of colorful fruits and vegetables to obtain essential nutrients naturally.

04. Hydration

Drinking enough fluids is extremely important.

Water helps:

- Prevent dehydration
- Support kidney function
- Reduce constipation

- **Replace fluids lost from vomiting or diarrhea**
- **Maintain normal body temperature**

Recommended fluids include:

- **Water**
 - **Coconut water**
 - **Clear soups**
 - **Fresh fruit juices (without added sugar)**
 - **Oral rehydration solutions when advised**
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05. Foods to Include

A healthy cancer diet should contain:

Fruits

Examples:

- **Apples**
 - **Bananas**
 - **Oranges**
 - **Papaya**
 - **Berries**
 - **Pomegranate**
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Vegetables

Examples:

- **Spinach**
- **Broccoli**
- **Carrots**
- **Tomatoes**
- **Cabbage**
- **Pumpkin**

Aim to include vegetables in every meal.

Whole Grains

Choose:

- Brown rice
- Whole wheat chapati
- Oats
- Millets
- Quinoa

These provide fiber and long-lasting energy.

Dairy Products

Include if tolerated:

- Milk
- Yogurt
- Cheese
- Paneer

These provide calcium and protein.

06. Foods to Limit

Certain foods should be avoided or consumed in moderation.

Examples include:

- Fried foods
- Processed meats
- Sugary beverages
- Fast food
- Excessive sweets
- Alcohol
- Highly processed snacks

These foods may provide little nutritional value.

07. Managing Poor Appetite

Many patients experience reduced appetite during treatment.

Helpful strategies include:

- Eat small meals every 2–3 hours.
 - Eat favorite healthy foods.
 - Choose calorie-rich nutritious snacks.
 - Eat when appetite is strongest.
 - Avoid skipping meals.
 - Drink fluids between meals rather than during meals.
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08. Managing Nausea Through Diet

To reduce nausea:

- Eat dry foods such as crackers.
 - Avoid greasy foods.
 - Eat slowly.
 - Drink cold or room-temperature beverages.
 - Avoid strong food odors.
 - Eat smaller portions.
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09. Managing Mouth Sores

When mouth sores are present:

Choose:

- Soft foods
- Mashed potatoes
- Yogurt
- Smoothies
- Soups
- Scrambled eggs

Avoid:

- Spicy foods
 - Citrus fruits
 - Hard foods
 - Hot beverages
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10. Managing Diarrhea

During diarrhea:

Choose:

- Bananas
- Rice
- Applesauce
- Toast
- Boiled potatoes

Avoid:

- High-fat foods
- Spicy foods
- Excessive dairy products
- Caffeinated drinks

Drink extra fluids to prevent dehydration.

11. Managing Constipation

To reduce constipation:

- Increase fiber intake gradually.
 - Drink sufficient water.
 - Eat fruits and vegetables.
 - Include whole grains.
 - Stay physically active.
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12. Food Safety

Cancer treatment can weaken the immune system.

To reduce infection risk:

- Wash fruits and vegetables thoroughly.
- Cook meat completely.
- Avoid raw eggs.
- Avoid unpasteurized milk.
- Wash hands before eating.

- Store food properly.

Food safety is especially important during chemotherapy.

13. Nutrition Before Surgery

Before surgery, patients should:

- Eat enough protein.
- Stay hydrated.
- Follow pre-surgery fasting instructions.
- Avoid alcohol.
- Follow the surgeon's dietary recommendations.

Good nutrition supports faster recovery after surgery.

14. Nutrition After Surgery

After surgery:

- Begin with liquids if advised.
- Progress gradually to soft foods.
- Eat protein-rich meals.
- Drink adequate fluids.
- Avoid overeating.

Healing requires increased nutritional support.

15. Maintaining Healthy Body Weight

Both weight loss and excessive weight gain may affect recovery.

Patients should:

- Monitor body weight regularly.
 - Discuss significant changes with healthcare providers.
 - Follow individualized dietary advice.
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16. Physical Activity and Nutrition

Regular light exercise combined with healthy nutrition helps:

- Maintain muscle strength

- Improve appetite
- Reduce fatigue
- Improve digestion
- Support emotional well-being

Exercise should always follow medical advice.

17. Dietary Supplements

Not all patients require vitamin or herbal supplements.

Supplements should only be taken if recommended by healthcare professionals.

Some supplements may interfere with cancer treatment.

18. Consultation with a Dietitian

Registered dietitians can provide personalized meal plans based on:

- Cancer type
- Treatment method
- Weight
- Medical conditions
- Food preferences

Individualized nutrition improves treatment outcomes.

19. Healthy Eating Tips

Patients should:

- Eat a balanced diet.
- Include fruits and vegetables daily.
- Choose lean protein sources.
- Drink enough water.
- Limit processed foods.
- Avoid smoking and alcohol.
- Maintain regular meal timings.

Healthy eating supports both treatment and long-term recovery.

20. Key Points

- **Nutrition plays a vital role during cancer treatment.**
- **Protein supports healing and muscle maintenance.**
- **Adequate hydration prevents dehydration and supports recovery.**
- **Fruits, vegetables, whole grains, and lean proteins should be included daily.**
- **Safe food handling reduces infection risk.**
- **Diet should be adjusted to manage treatment side effects.**
- **Professional dietary guidance helps optimize nutritional status.**
- **A balanced diet improves quality of life and overall treatment success.**